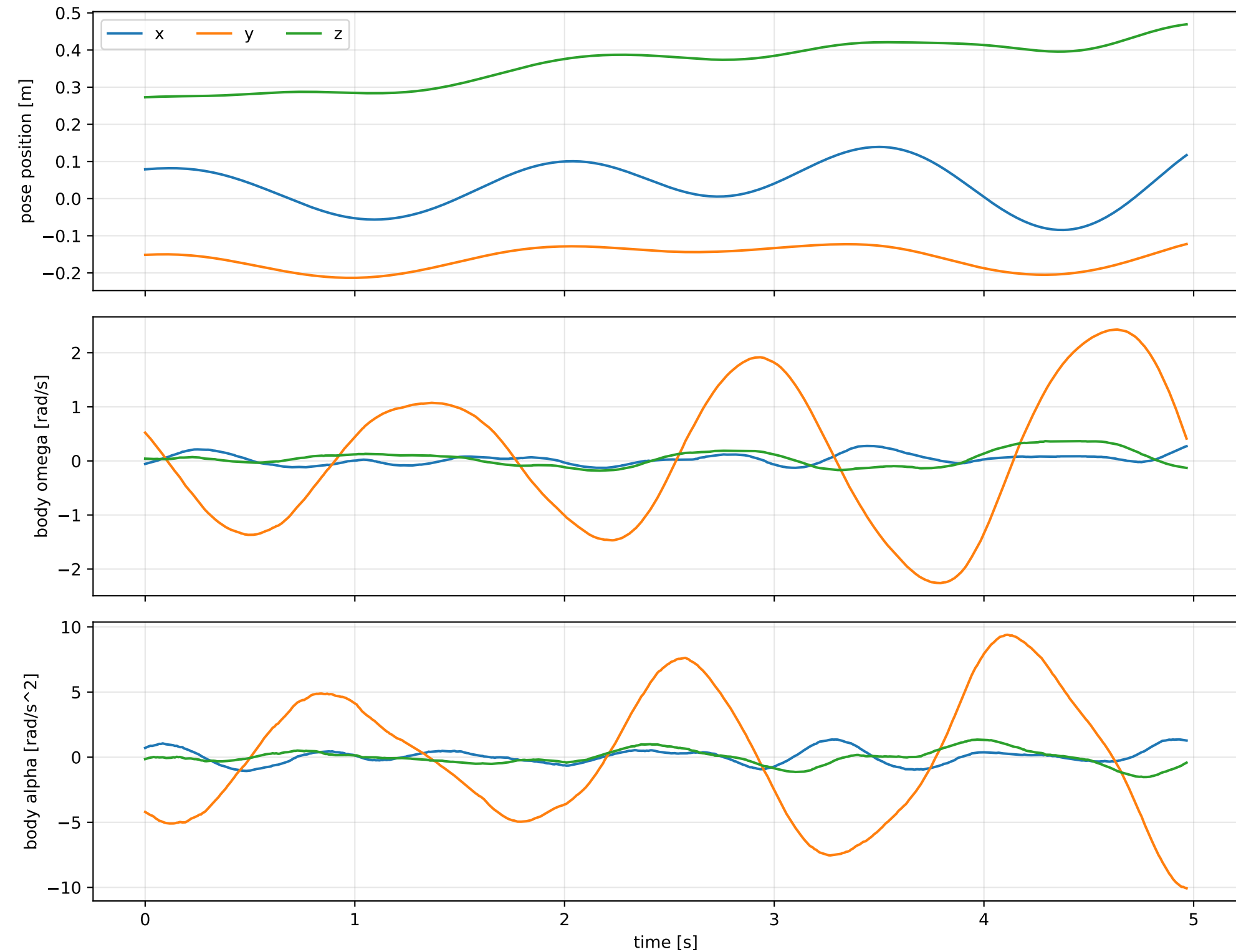
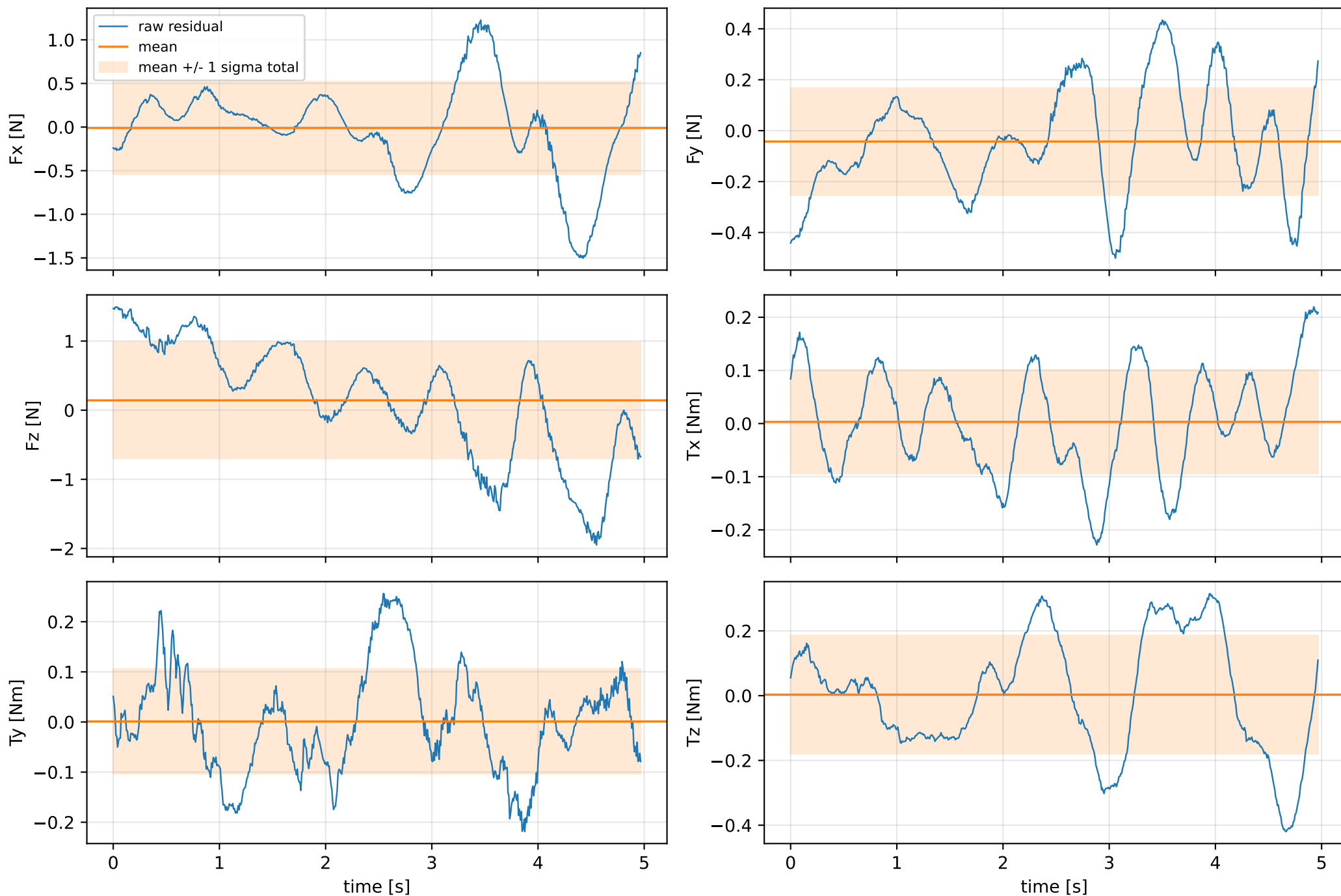


inertia_over_mass_xz_nominal: SG trajectory and derived motion



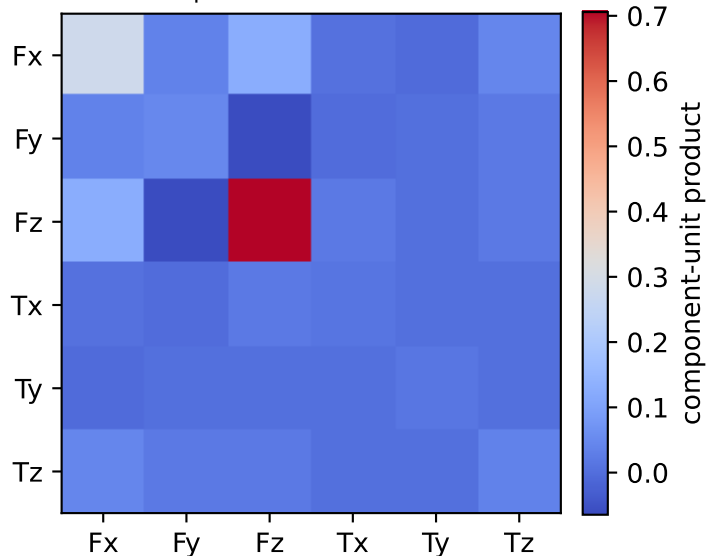
inertia_over_mass_xz_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



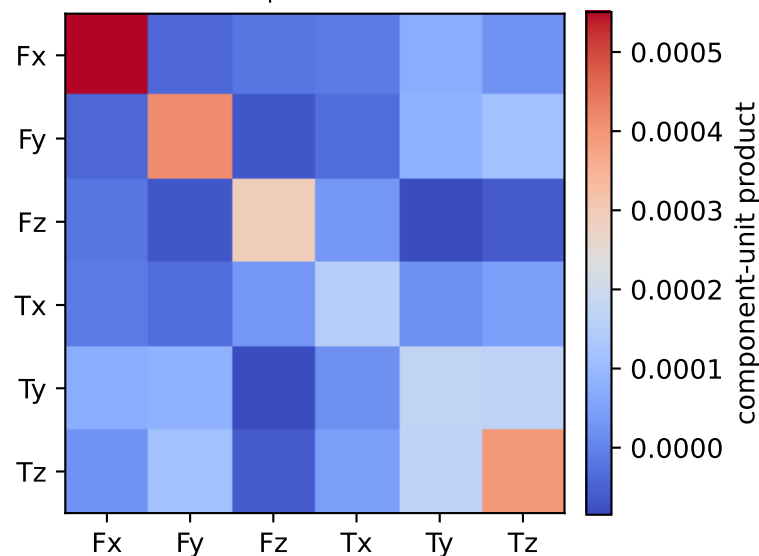
vector RMS about zero: force 1.0253 N, torque 0.23125 Nm; component std about mean: force [0.5302 0.2098 0.8401], torque [0.0972 0.1044 0.1822]

inertia_over_mass_xz_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.008 0.01 0.017 0.039 0.265 0.745]

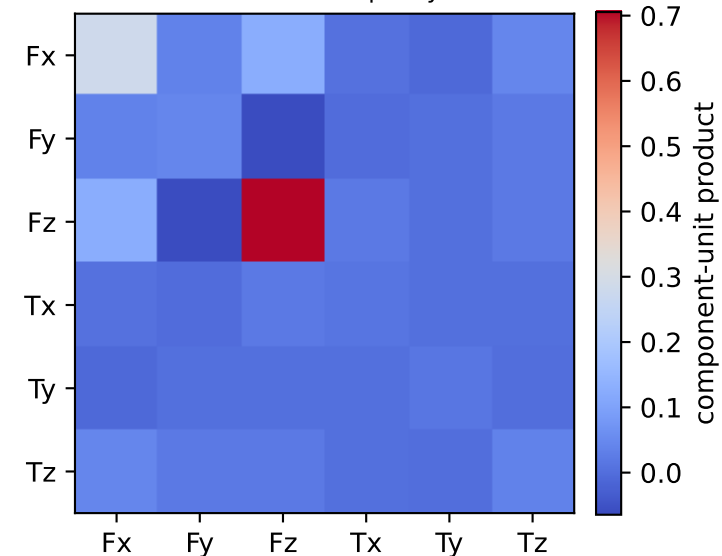
empirical total covariance



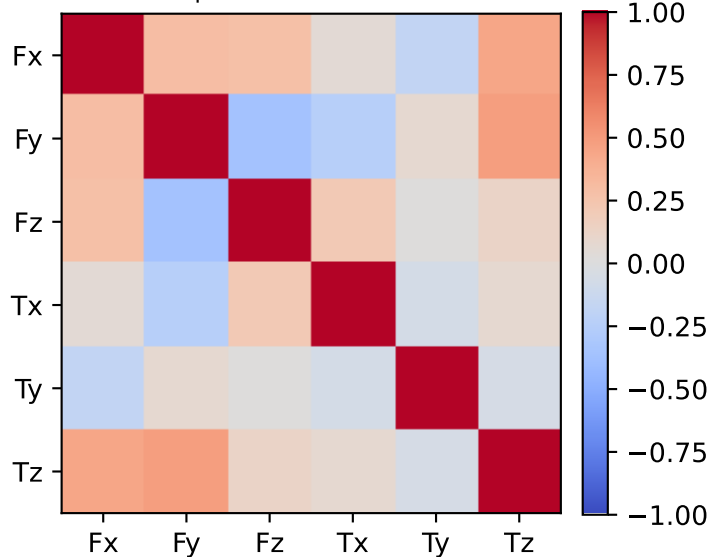
mean full-SG prediction covariance



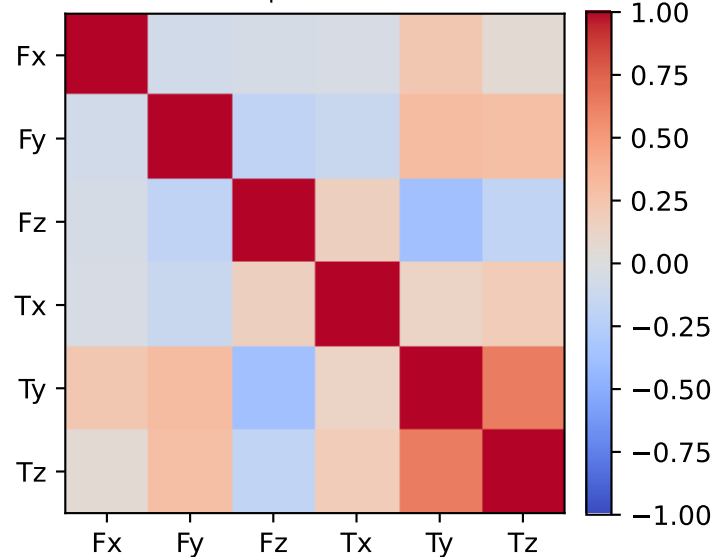
PSD excess model discrepancy covariance



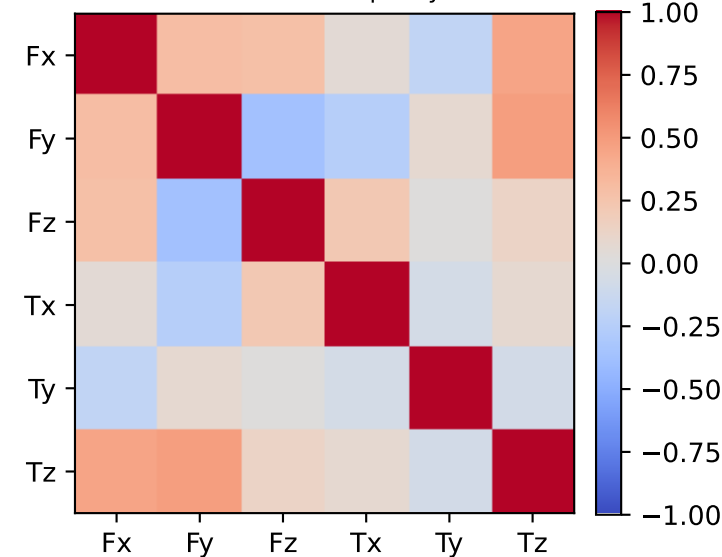
empirical total correlation



mean full-SG prediction correlation

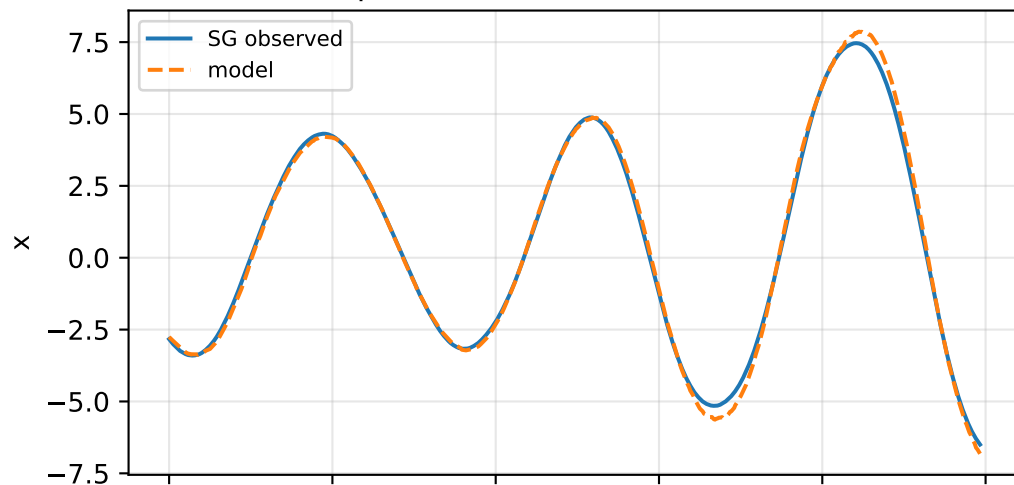


PSD excess model discrepancy correlation

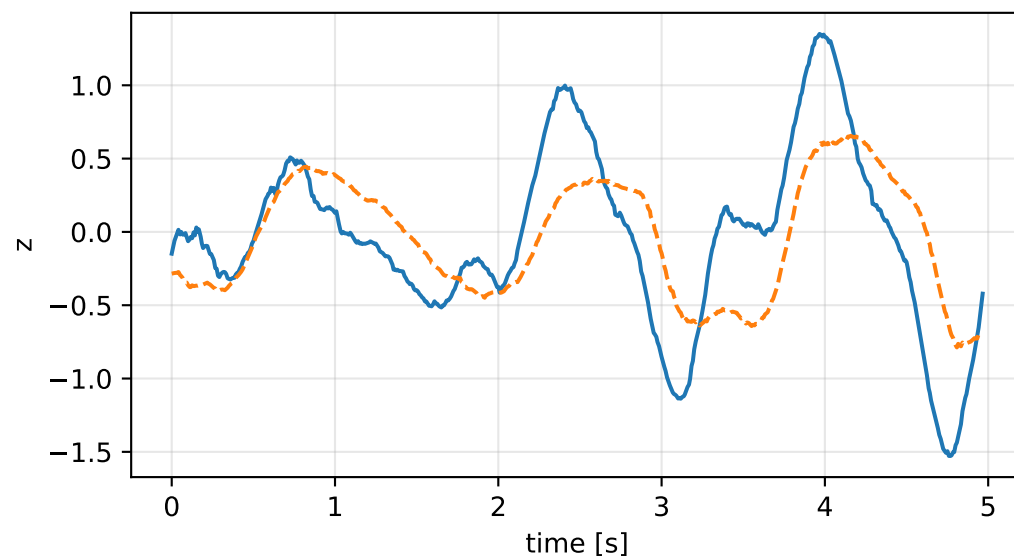
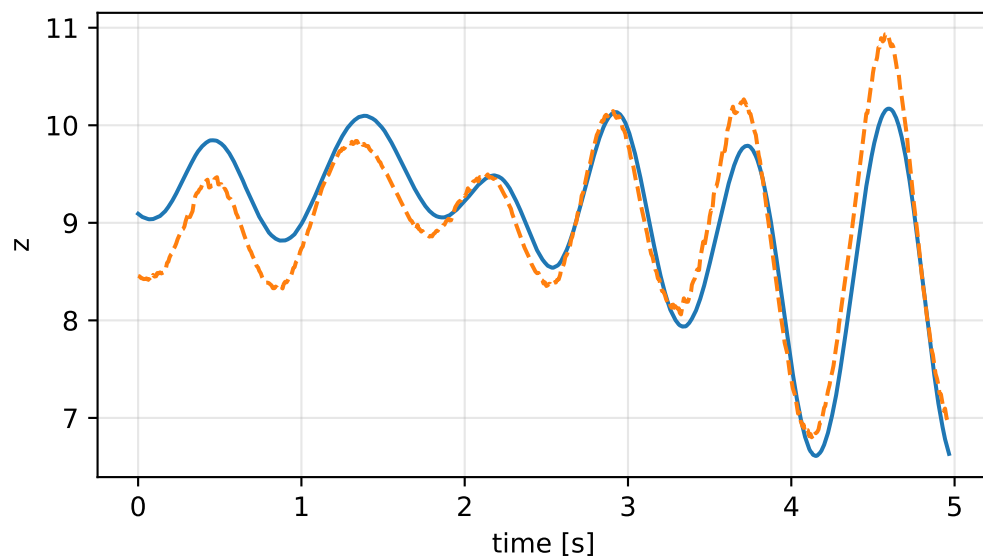
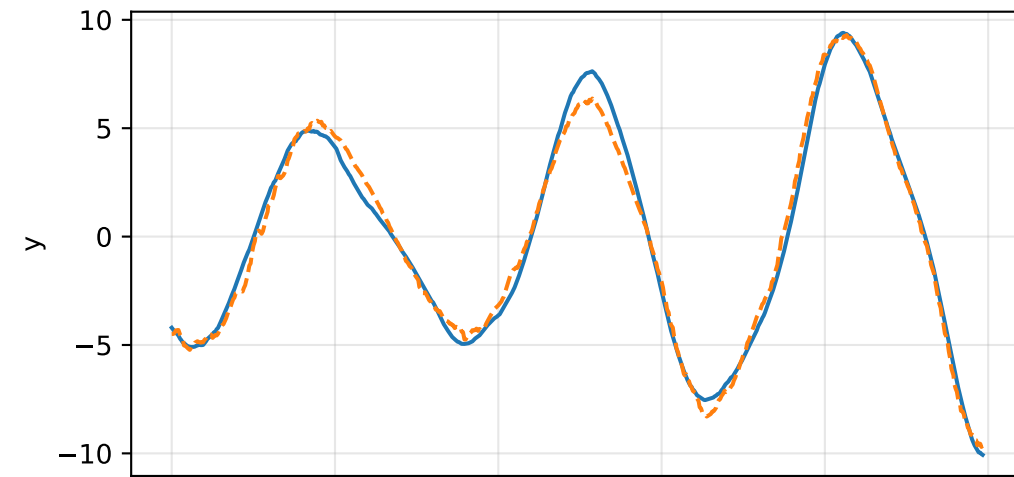
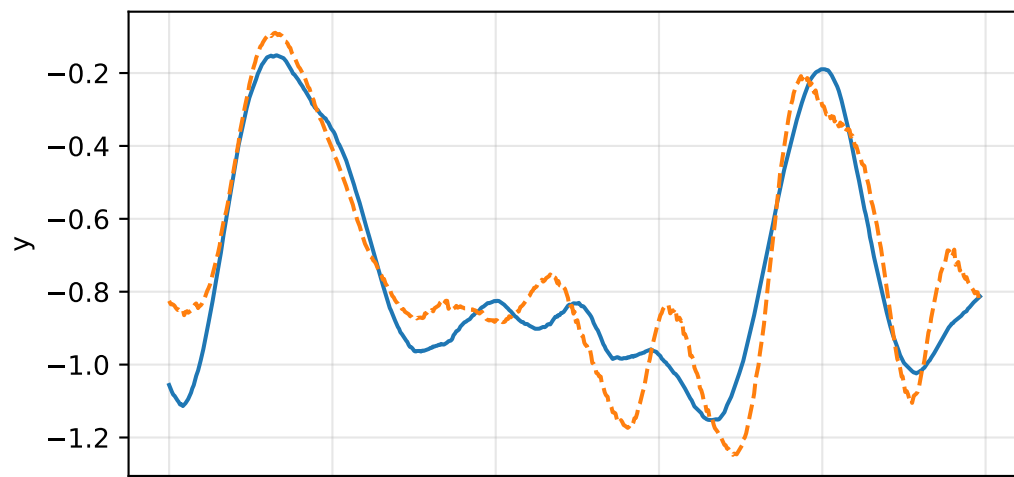
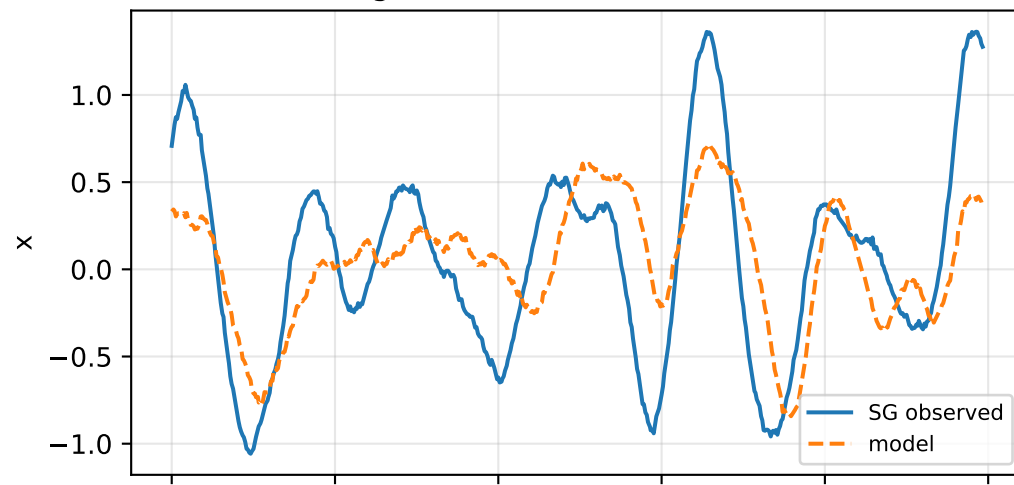


inertia_over_mass_xz_nominal: acceleration residual objective

specific acceleration [m/s^2]

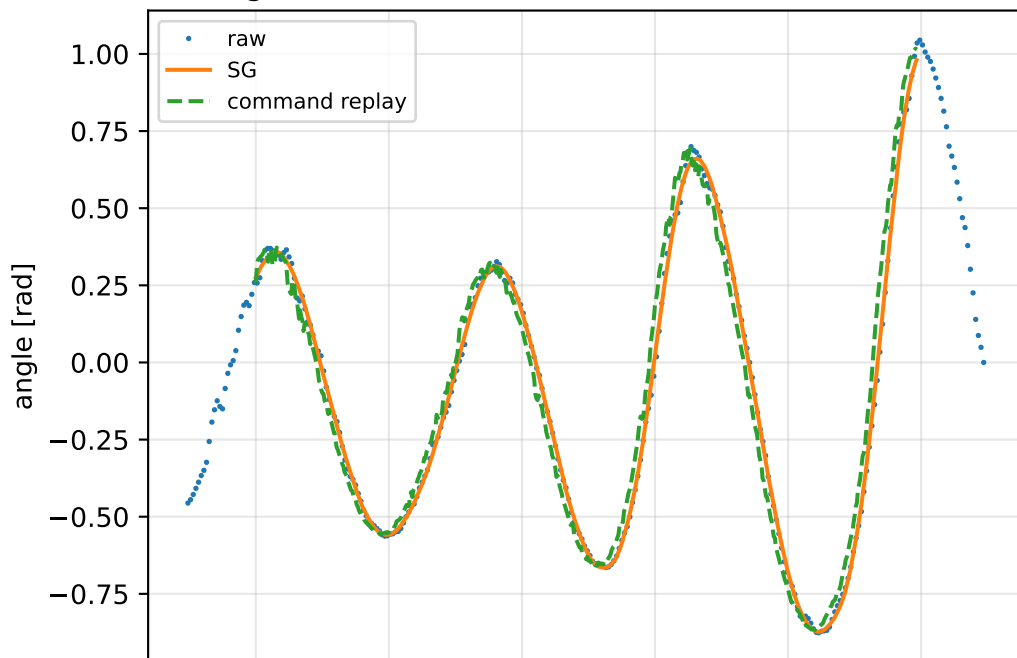


angular acceleration [rad/s^2]

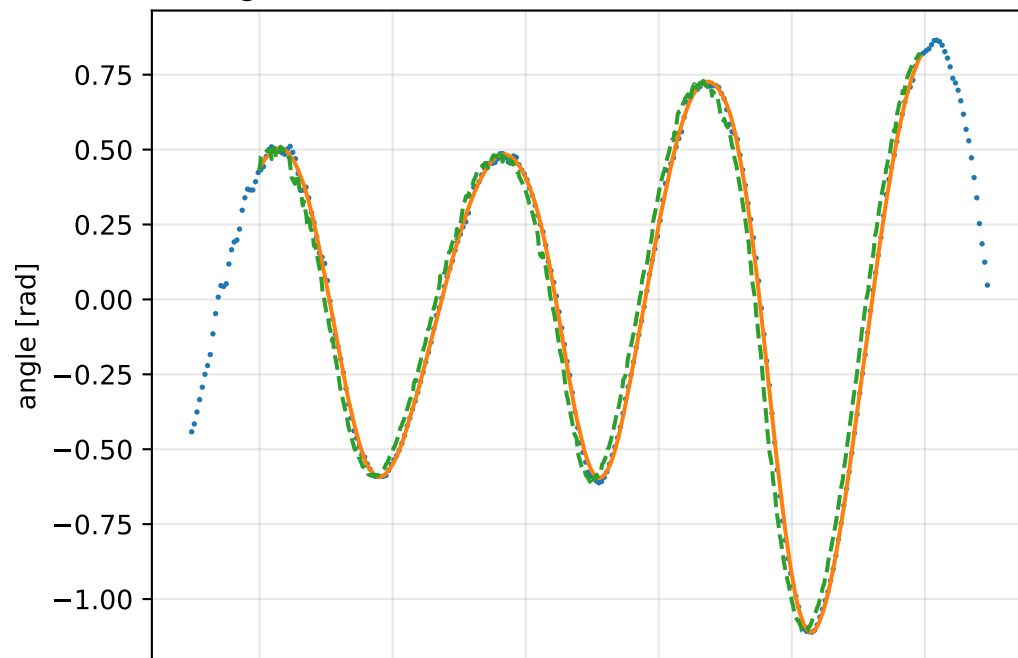


inertia_over_mass_xz_nominal: gimbal measurement audit (objective=measured_sg)

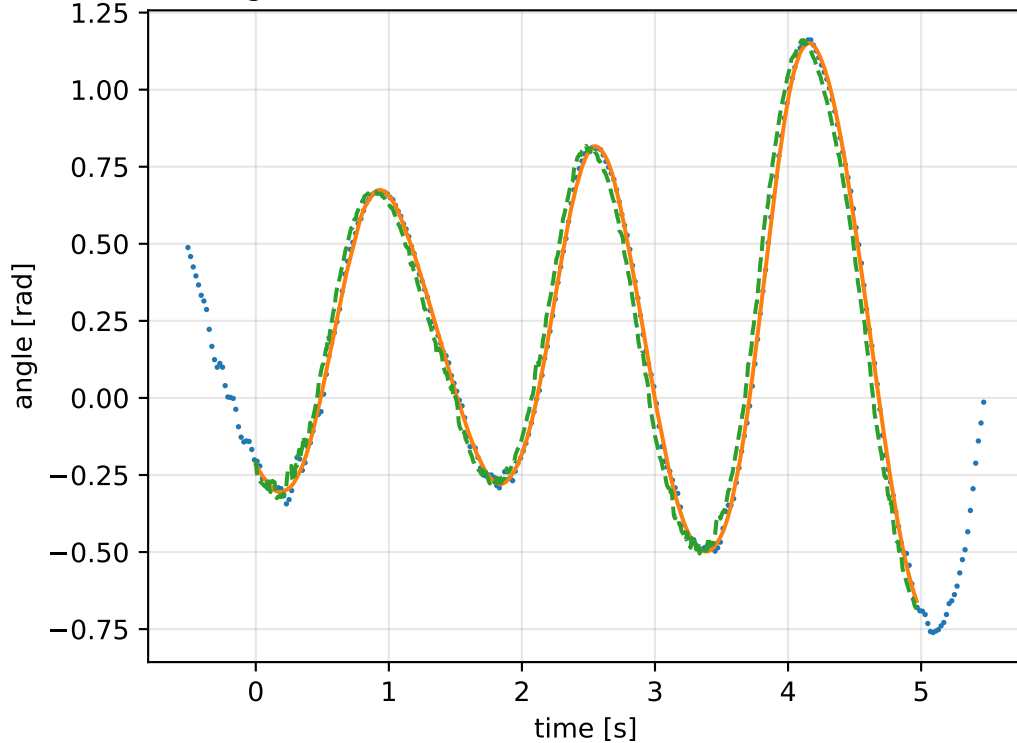
gimbal 1: RMSE=0.0759, max=0.2095 rad



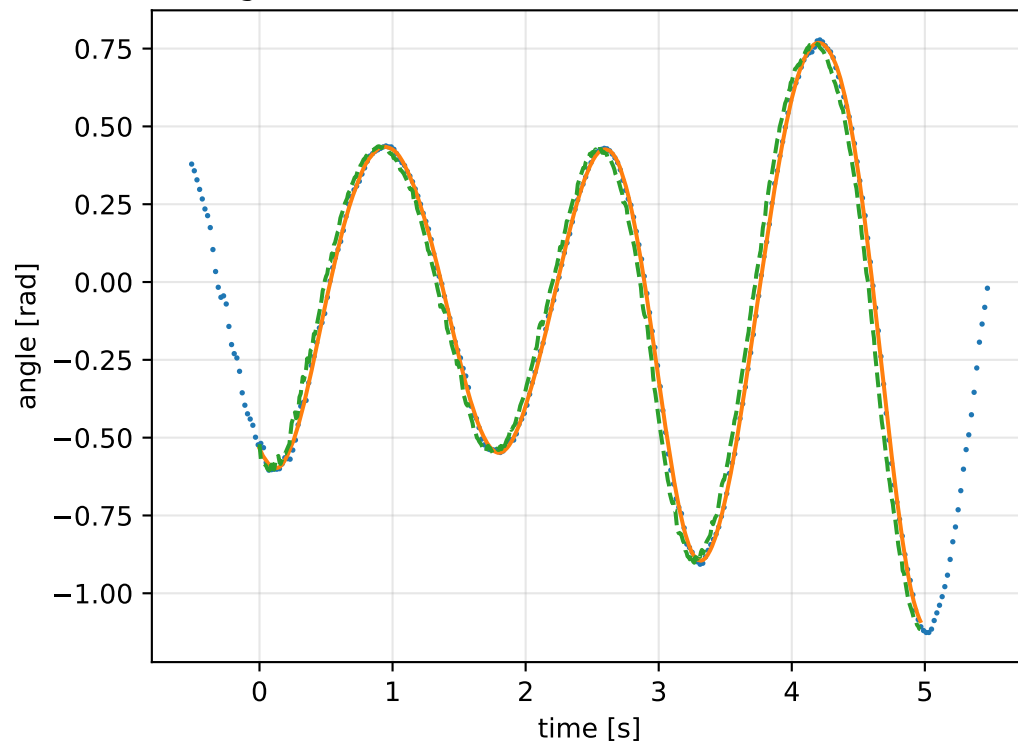
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

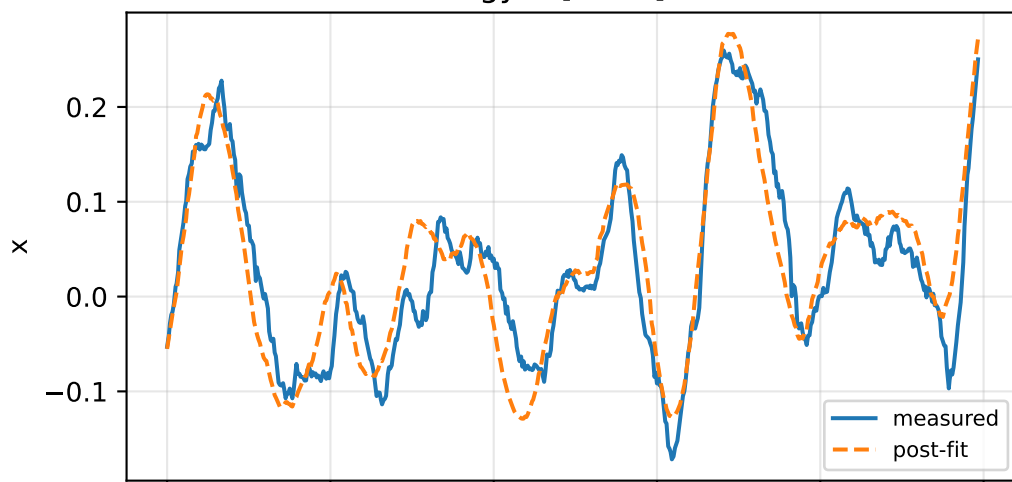


gimbal 4: RMSE=0.07131, max=0.1737 rad

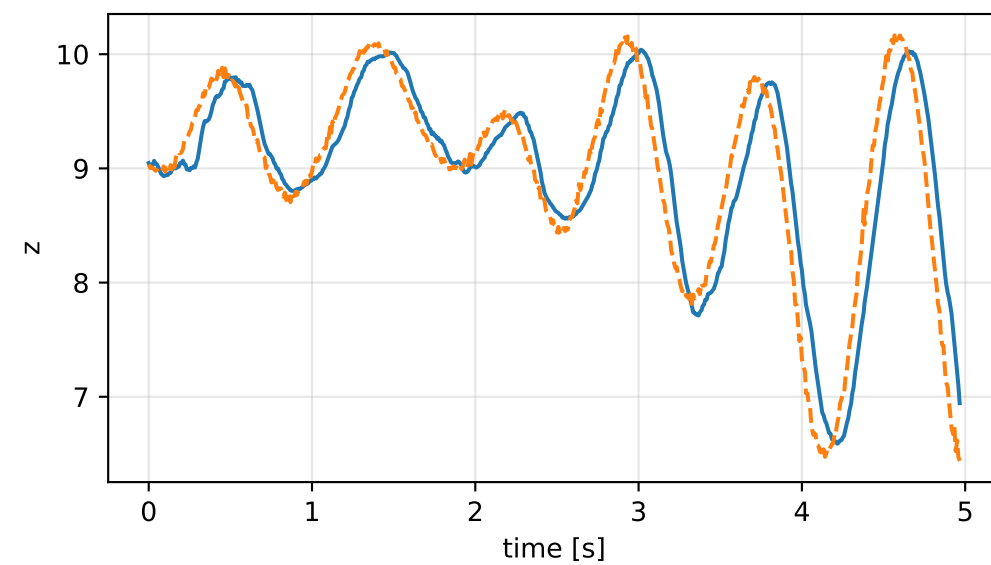
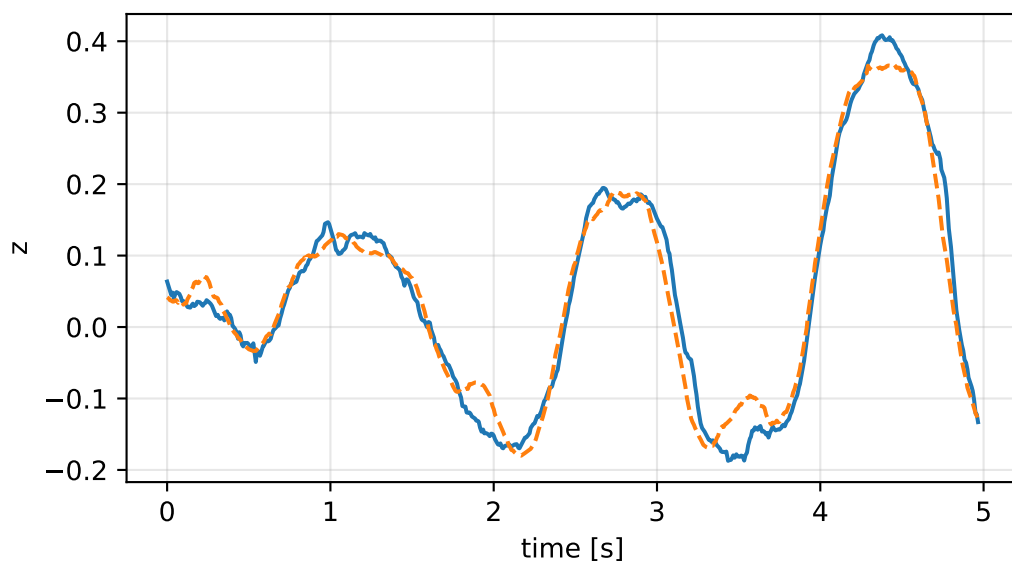
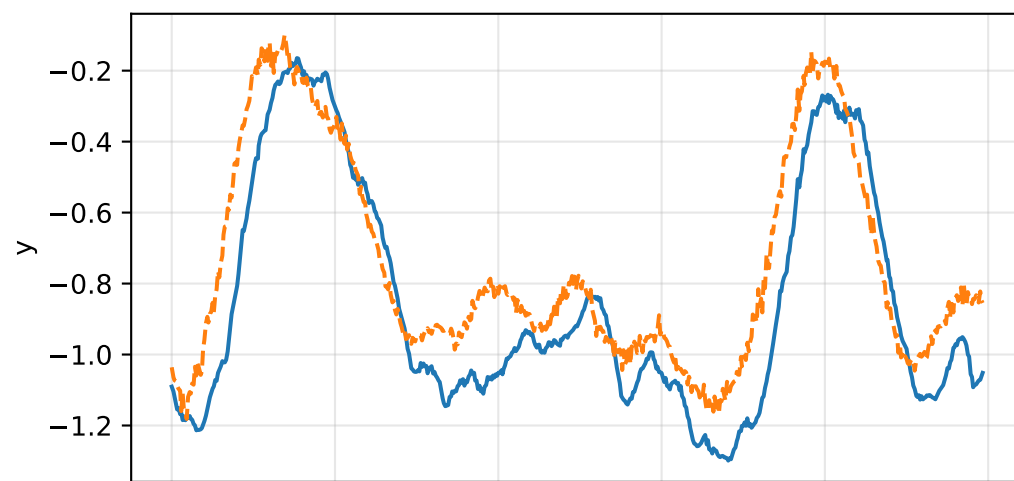
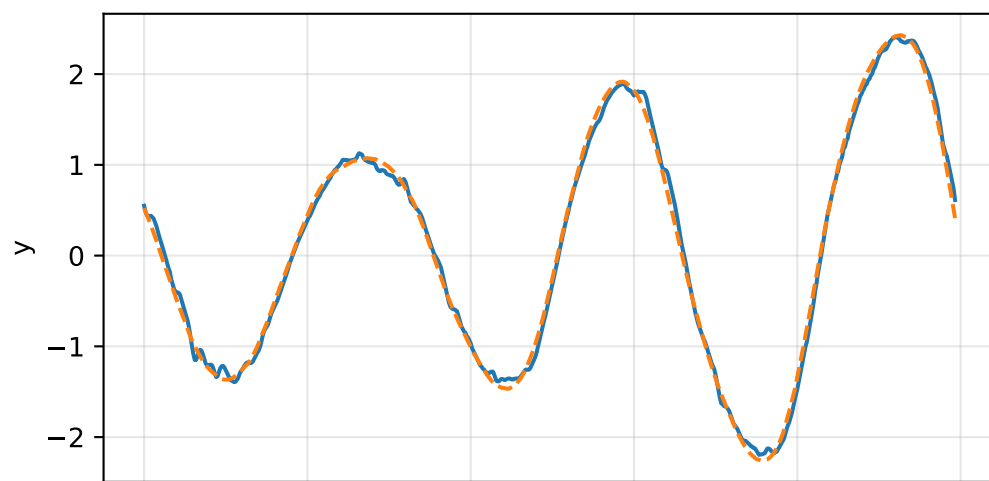
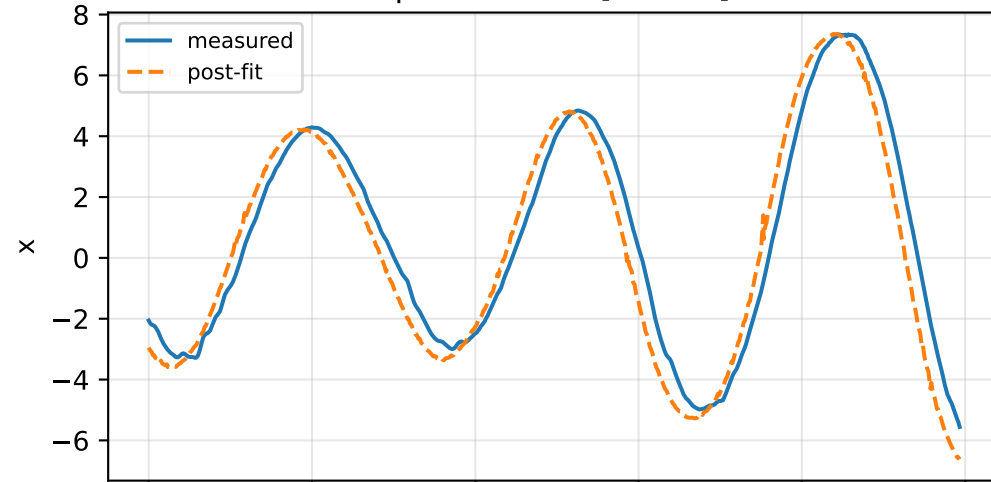


inertia_over_mass_xz_nominal: IMU comparison (diagnostic only)

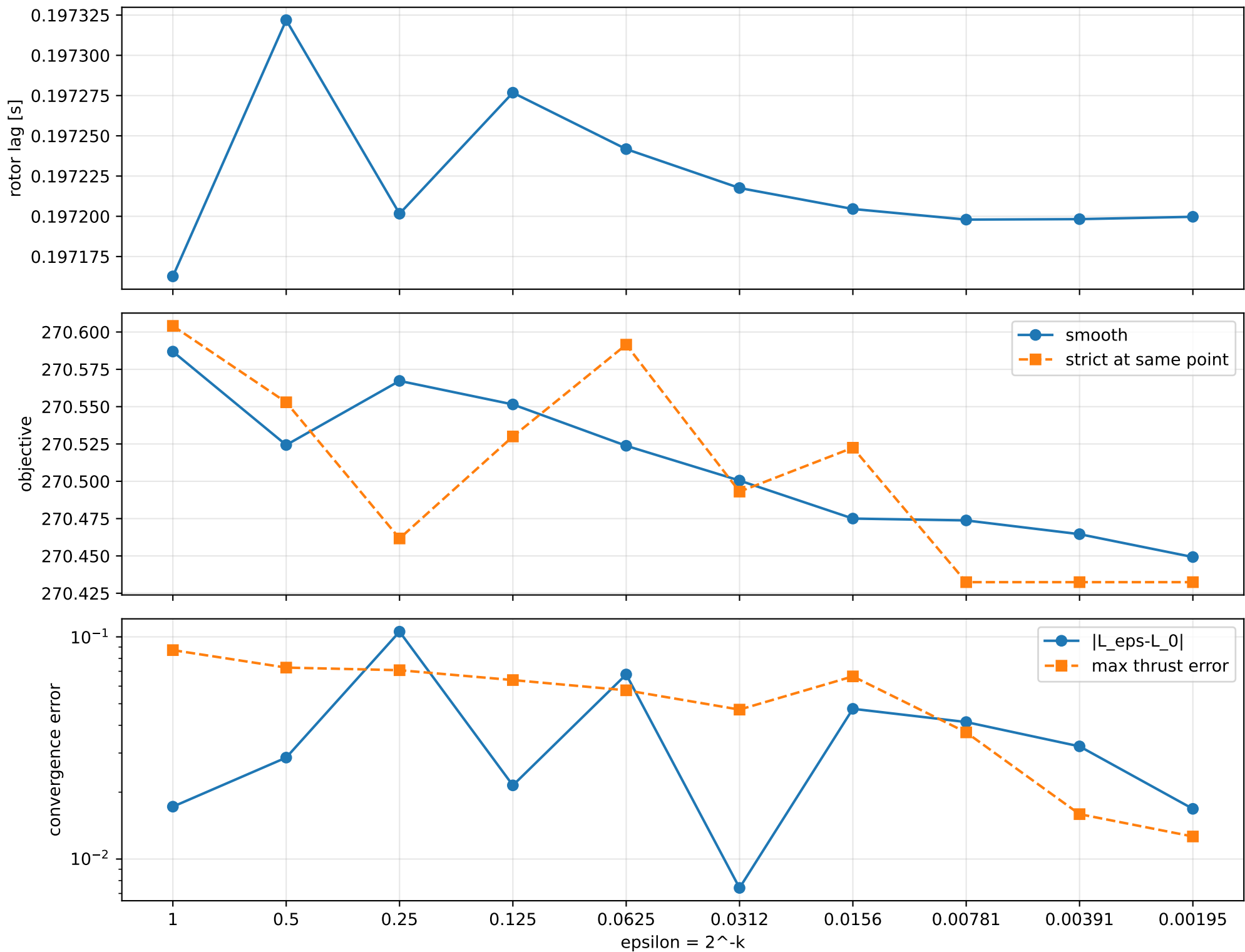
gyro [rad/s]



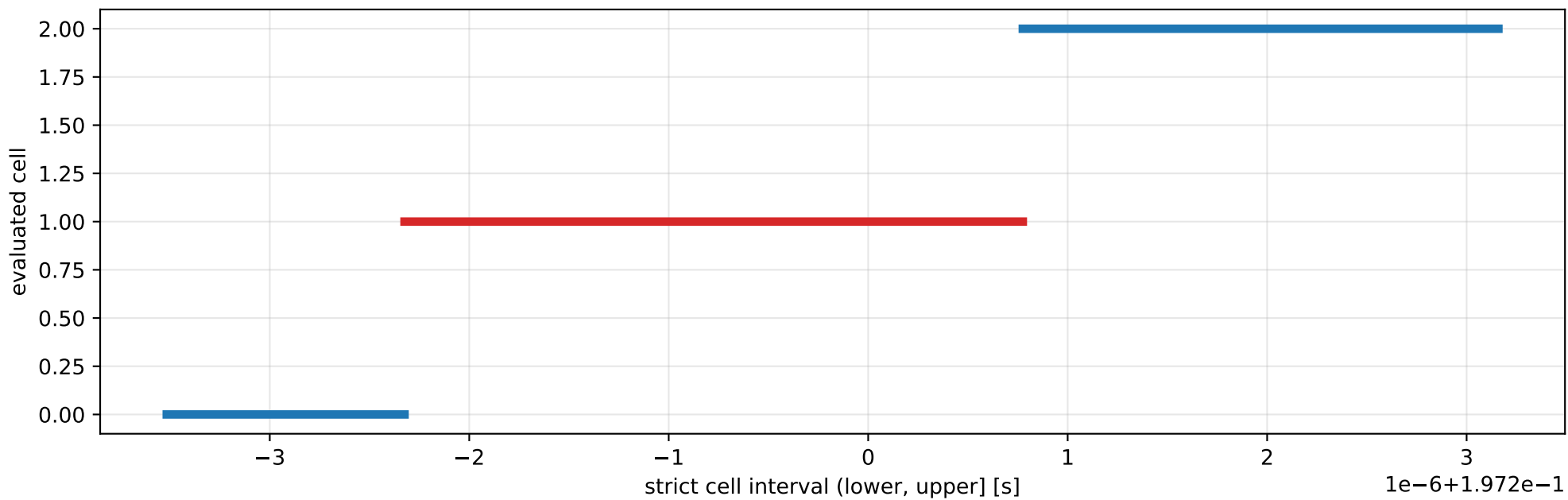
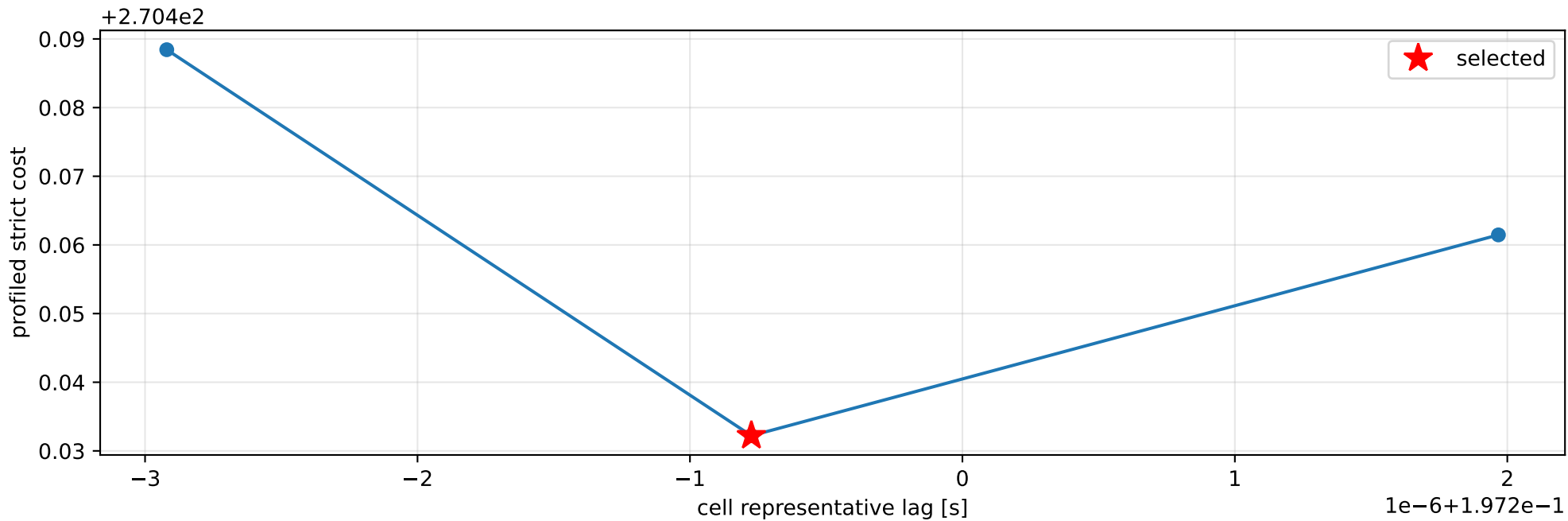
specific force [m/s^2]



inertia_over_mass_xz_nominal: smooth-to-strict continuation



inertia_over_mass_xz_nominal: exact strict-ZOH lag cells



selected (0.197197676, 0.197200775] s; final smooth 0.197199697 s; data boundary=False

inertia_over_mass_xz_nominal: parameter estimate

Case: inertia_over_mass_xz_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 12.942647

inertia [kg m²]:

```
[[ 1.24464169e+00 -2.05208554e-02  1.04651749e-04]
 [-2.05208554e-02  9.98408934e-01 -5.42066732e-03]
 [ 1.04651749e-04 -5.42066732e-03  2.24300333e+00]]
```

CoG body [m]: [0.01982813 -0.0034507 -0.0442528]

force effectiveness: [4.29398435 5.59896488 4.95001432 3.99154852]

scale-free J/m [m²]:

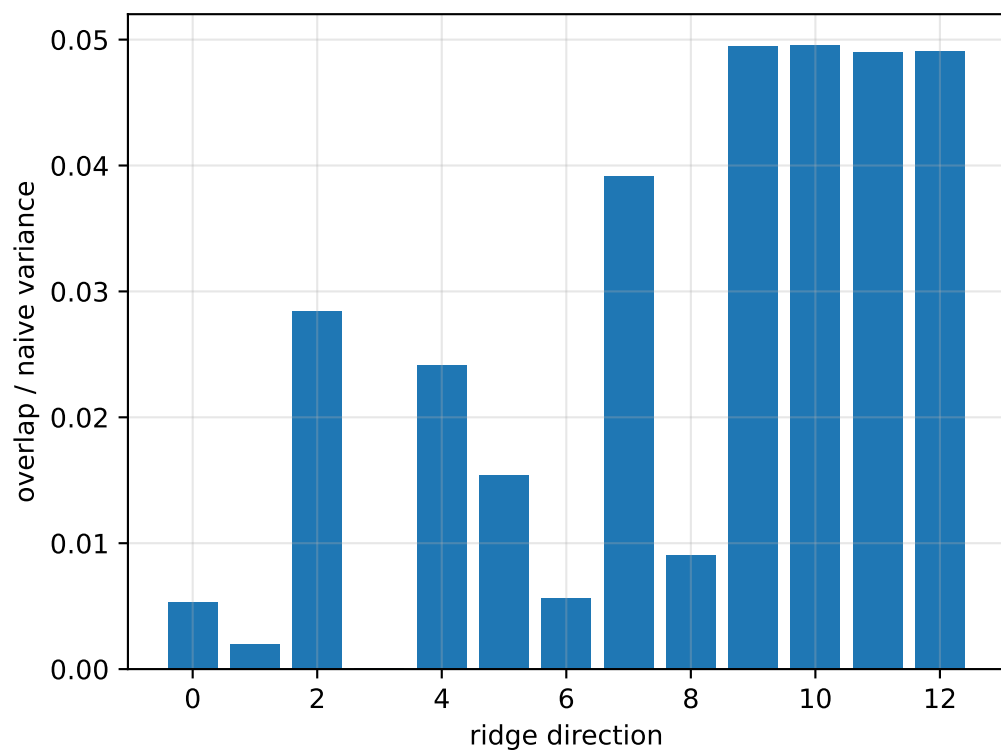
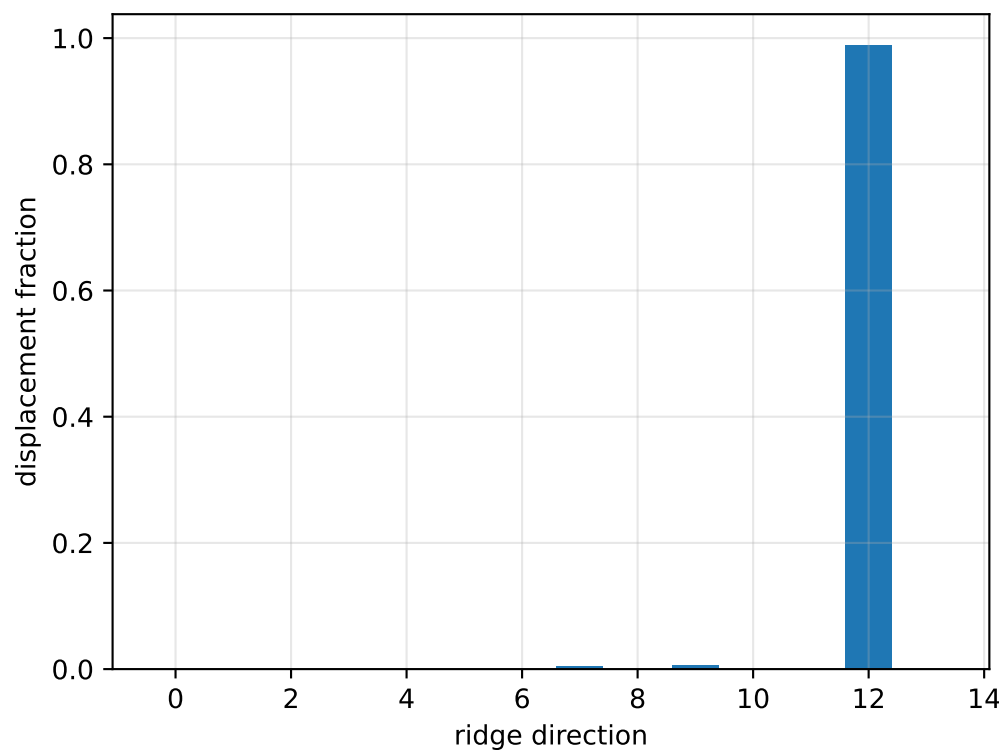
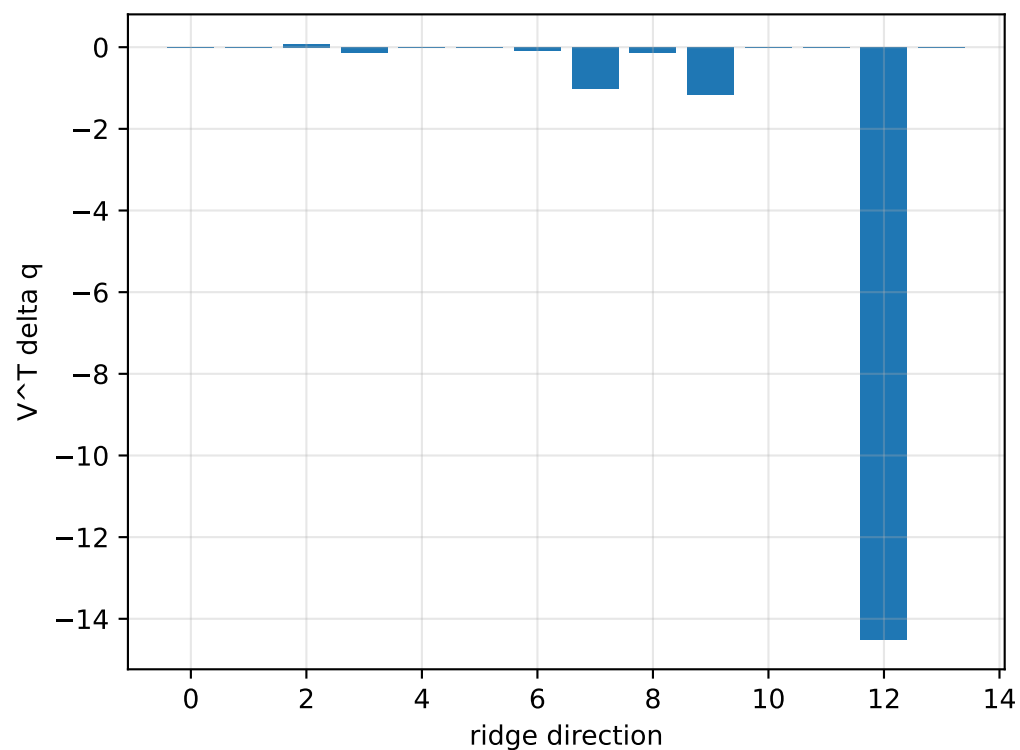
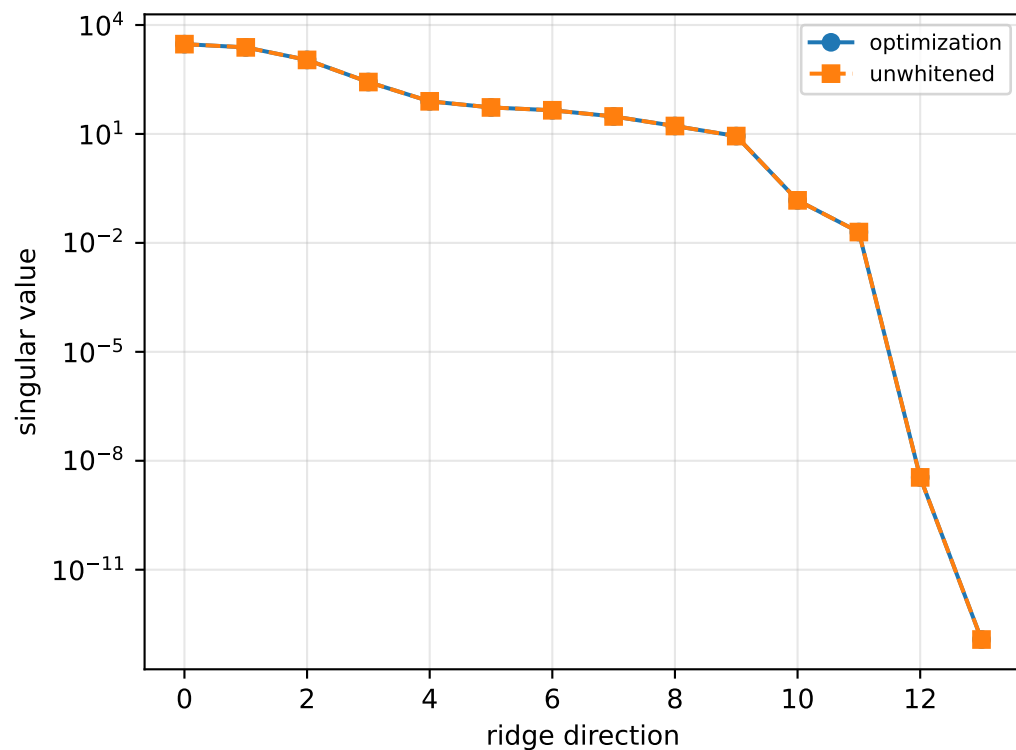
```
[[ 9.61659300e-02 -1.58552229e-03  8.08580724e-06]
 [-1.58552229e-03  7.71410155e-02 -4.18822156e-04]
 [ 8.08580724e-06 -4.18822156e-04  1.73303291e-01]]
```

scale-free f/m: [0.33177018 0.43259813 0.38245765 0.3084028]

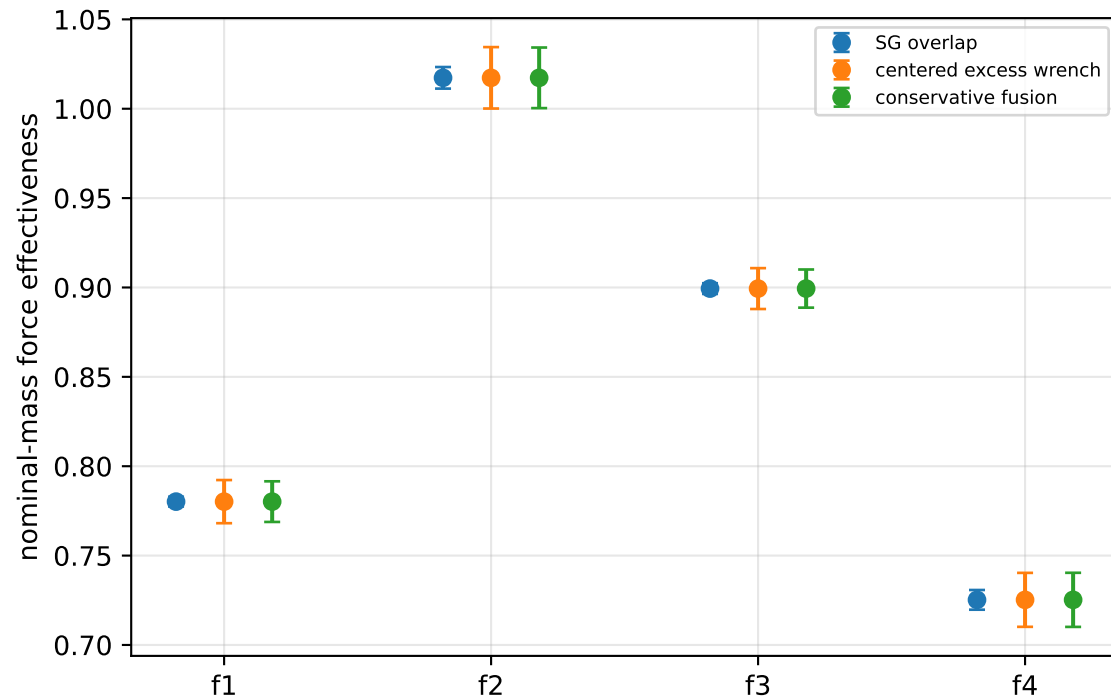
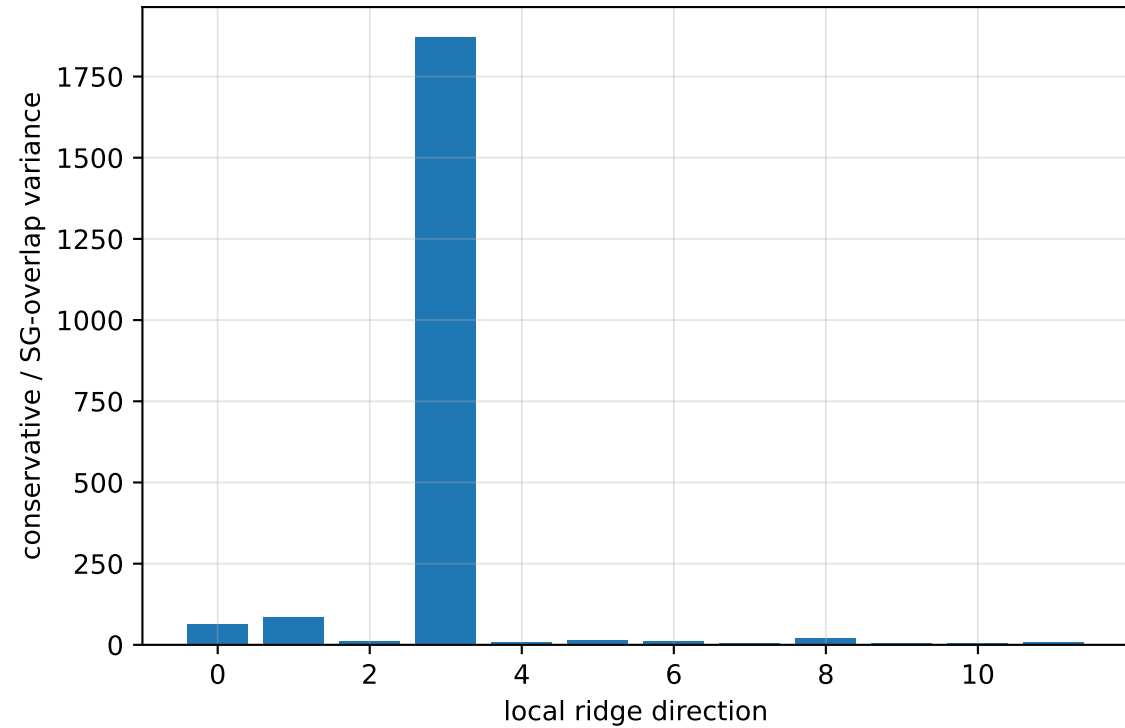
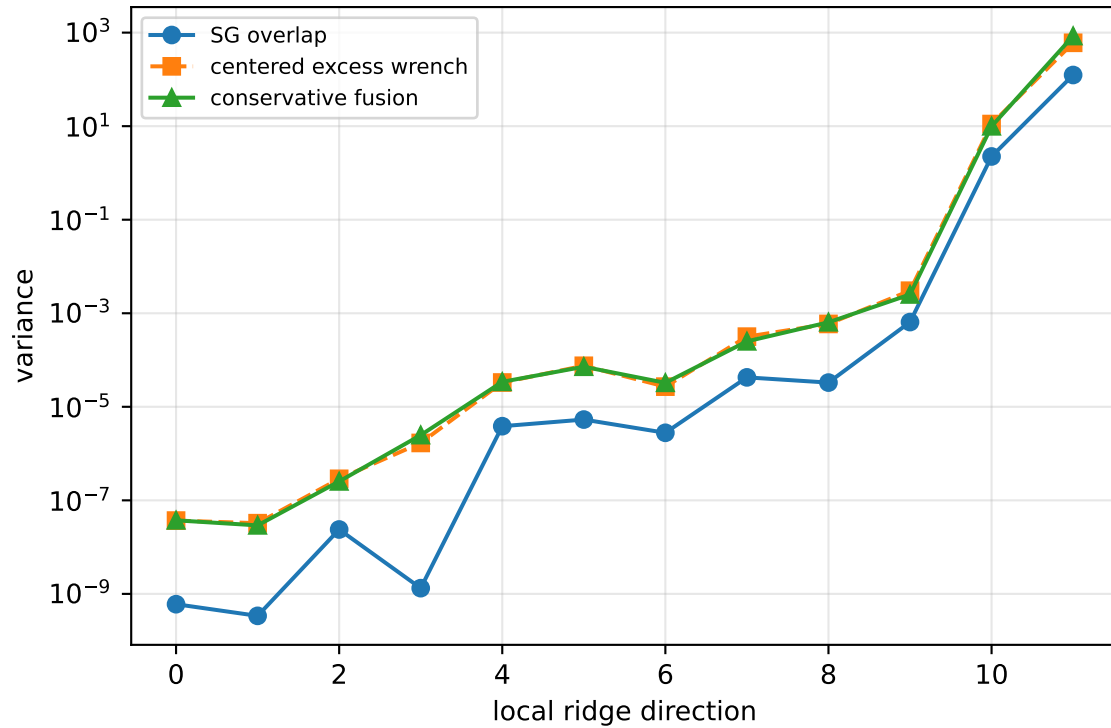
rotor lag cell: (0.197197676, 0.197200775] s

representative: 0.197199225 s

inertia_over_mass_xz_nominal: ridge spectrum (rank 13, nullity 1, ||v||=8.96e-14)



inertia_over_mass_xz_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions

$\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 45.28$

wrench-to-acceleration recovery max error = 3.47×10^{-15}

exact scale gauge ridge index = 13 (alignment 1)

index 11: variance=844.4, ratio=6.803

$\log_{\text{mass}} = +3.77 \times 10^{-5}$; second_moment_diag_1=+0.00126

second_moment_diag_2=-0.00124; second_moment_diag_3=-0.00021

second_moment_offdiag_12=+0.0481; second_moment_offdiag_13=-0.999

second_moment_offdiag_23=+0.0147; cog_x=+1.42e-06

cog_y=+1.39e-05; cog_z=-4.8e-06

$\log_{\text{force_eff_1}} = +4.4 \times 10^{-6}$; $\log_{\text{force_eff_2}} = -2.94 \times 10^{-5}$

$\log_{\text{force_eff_3}} = +0.000126$; $\log_{\text{force_eff_4}} = +5.55 \times 10^{-5}$

index 10: variance=9.921, ratio=4.41

$\log_{\text{mass}} = -0.000654$; second_moment_diag_1=+0.000506

second_moment_diag_2=-0.0693; second_moment_diag_3=+0.0713

second_moment_offdiag_12=-0.00197; second_moment_offdiag_13=+0.0146

second_moment_offdiag_23=+0.995; cog_x=-0.000266

cog_y=+0.000625; cog_z=-5.94e-05

$\log_{\text{force_eff_1}} = -0.00202$; $\log_{\text{force_eff_2}} = -0.00385$

$\log_{\text{force_eff_3}} = +0.00058$; $\log_{\text{force_eff_4}} = +0.00349$

index 9: variance=0.002519, ratio=3.888

$\log_{\text{mass}} = +0.143$; second_moment_diag_1=+0.126

second_moment_diag_2=-0.89; second_moment_diag_3=-0.00221

second_moment_offdiag_12=+0.00402; second_moment_offdiag_13=+0.000569

second_moment_offdiag_23=-0.0627; cog_x=-0.0263

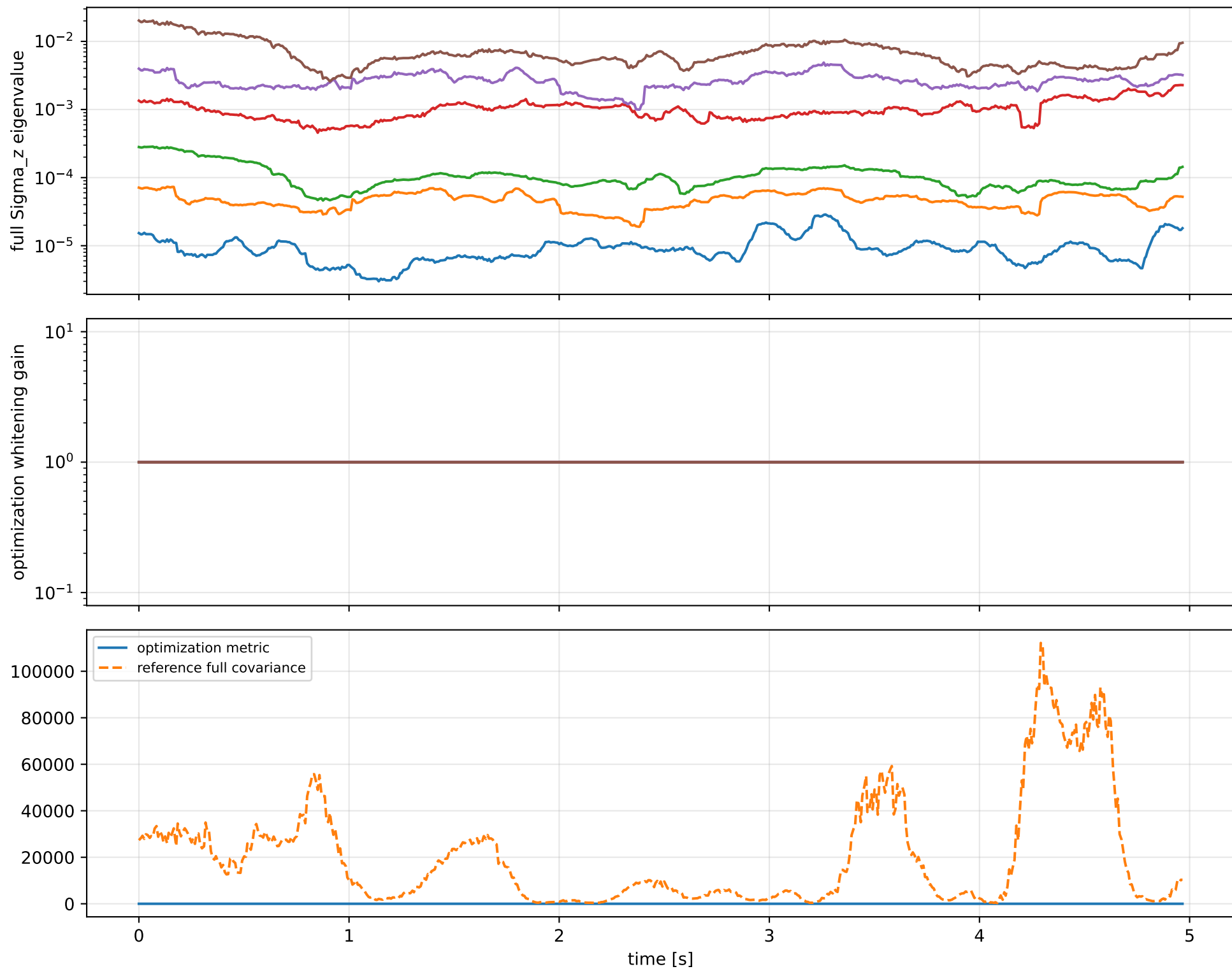
cog_y=+0.0123; cog_z=+0.0042

$\log_{\text{force_eff_1}} = +0.209$; $\log_{\text{force_eff_2}} = +0.00358$

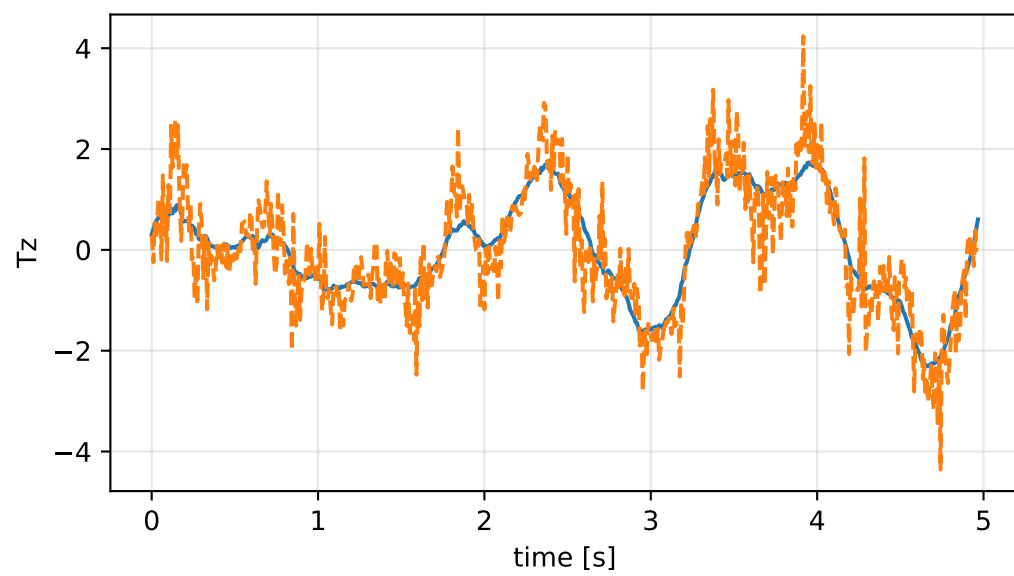
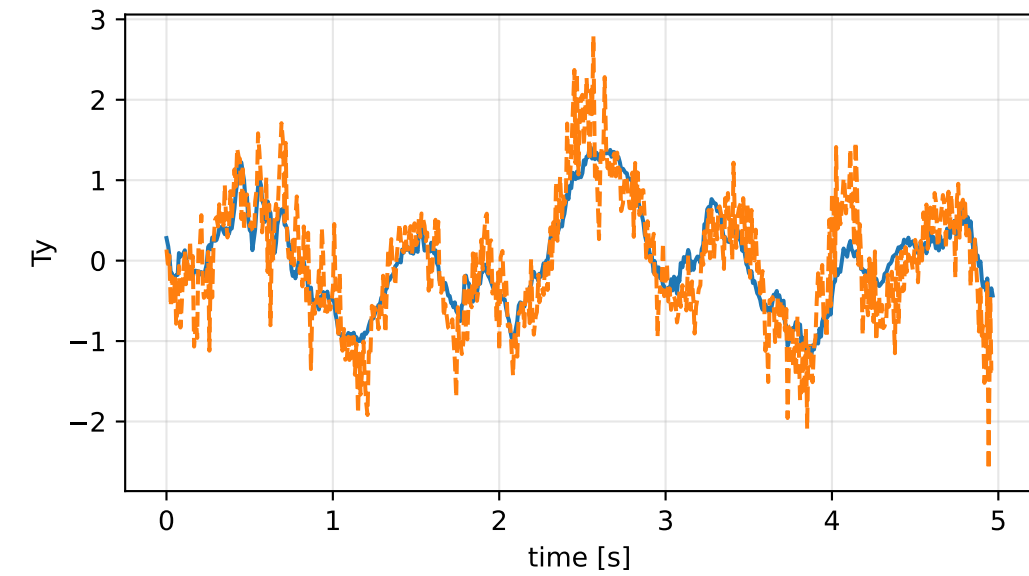
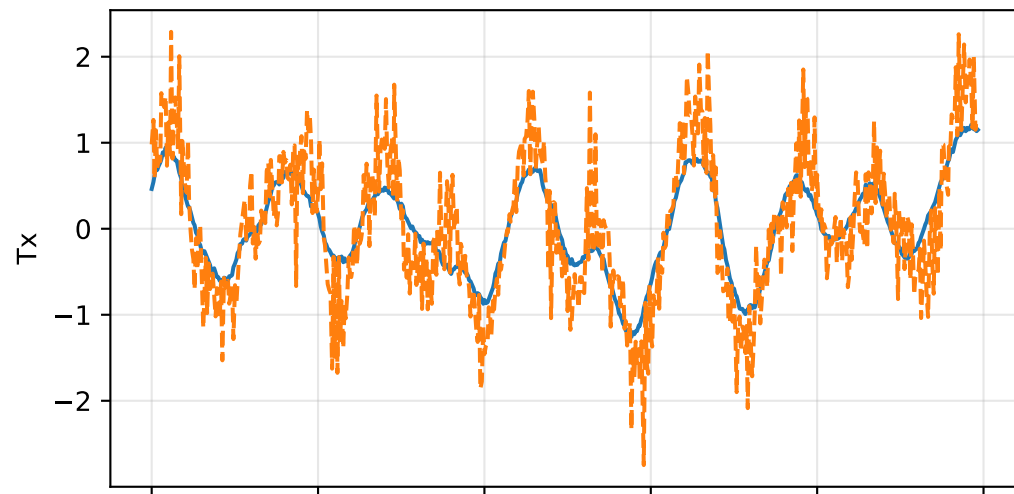
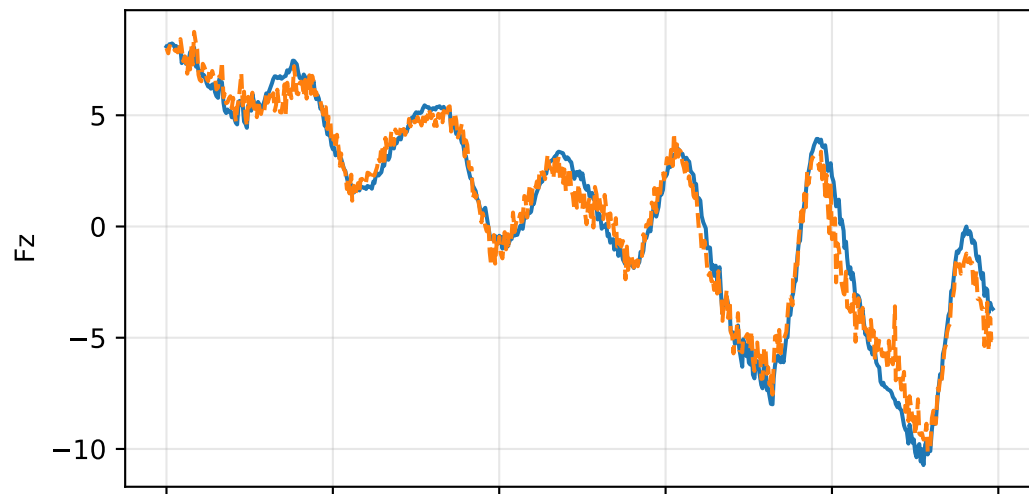
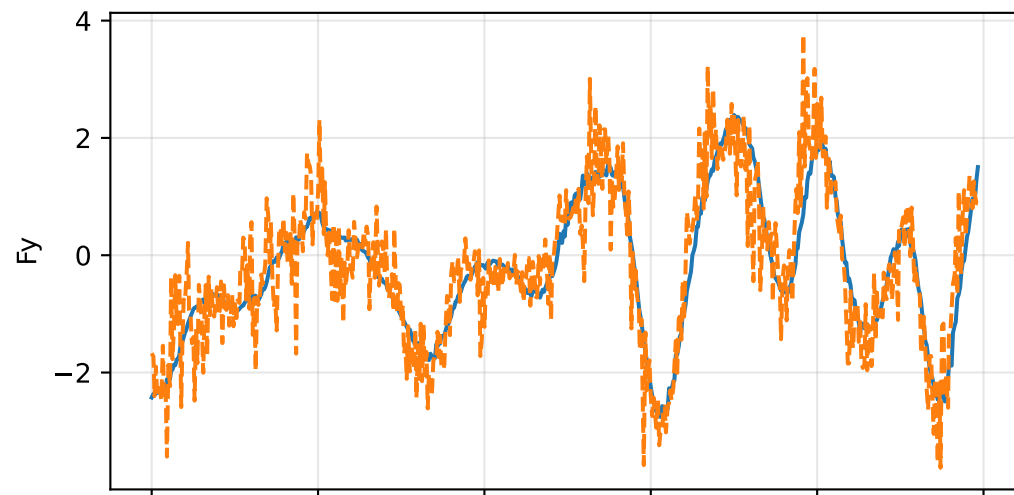
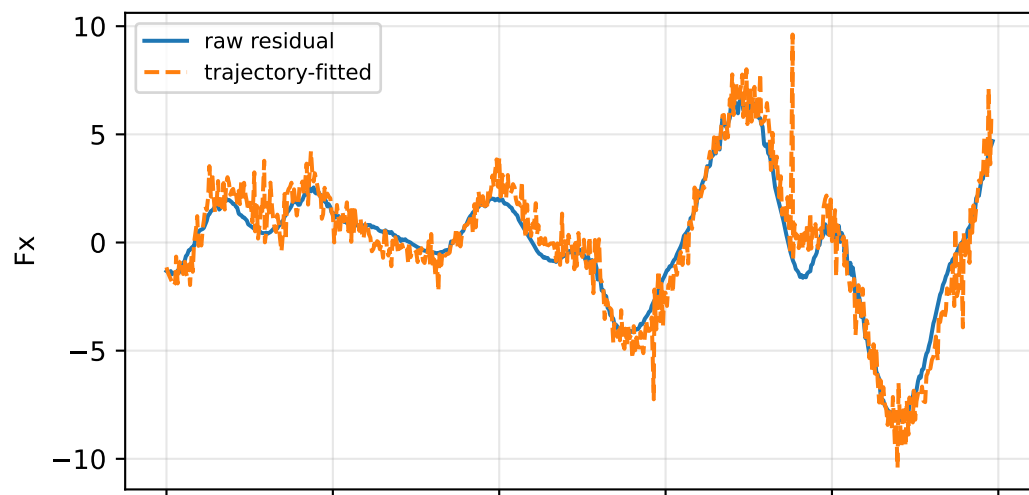
$\log_{\text{force_eff_3}} = +0.0679$; $\log_{\text{force_eff_4}} = +0.343$

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

inertia_over_mass_xz_nominal: optimization vs reference covariance



inertia_over_mass_xz_nominal: wrench / closure (force max 3.55e-14, torque max 3.47e-15)



inertia_over_mass_xz_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_inertia_over_mass_xz_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/inertia_over_ma

SHA256: 99d969ed0e0fa85ae900b089864493fcb7e345fdd5fd63009a6ebdbc6ee04171

data objective: 270.432214804

prior objective: 2.40002389854e-08

total objective: 270.432214828

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: inertia_over_mass_xz_nominal (inertia_over_mass_m2)

components: xz

target: [8.08602633e-06]

std: [1.e-06]

final: [8.08580724e-06]

error: [-2.19090114e-10]

standardized residual: [-0.00021909]

factor objective: 2.40002389854e-08